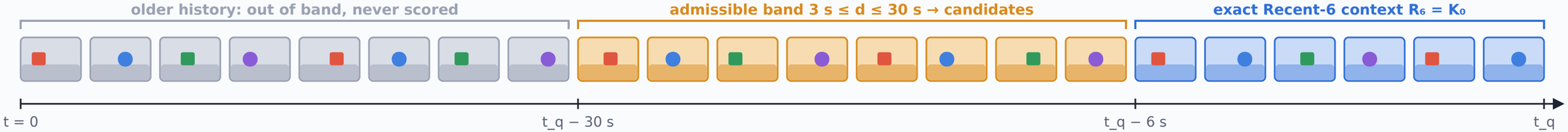
Streaming input (1 fps) *the frozen VLM only ever sees the exact Recent-6 window plus at most 3 admitted history frames*

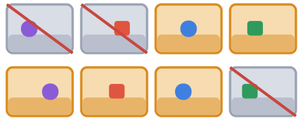


Question q at t_q

"What did the person put on the table?"

open-ended or multiple-choice: one controller, no option scores

1 Temporal pre-filter



keep h only if

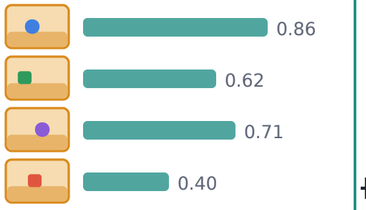
$d_{min} \le t_s(R_6) - t_e(h) \le d_{max}$

(3 s, 30 s); horizon 64

drops near-duplicates of R_6 and stale evidence before any retrieval cost is paid

→ candidate set H_t

2 CLIP-MMR retrieval ❄



query = question text only

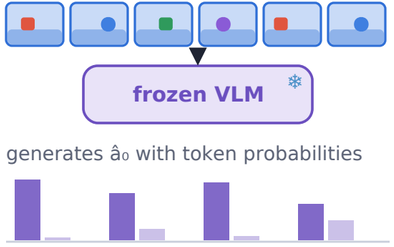
$r_i = \cos(E_v(h_i), E_t(q))$

MMR: $\lambda\, r_i - (1-\lambda) \max_s \mathrm{sim}(h_i, h_s)$

λ = 0.8 → diverse, relevant queue

h₁ h₂ h₃ ranked

3 Recent sufficiency ($K_0 = R_6$)



generates $\hat{a}_0$ with token probabilities

$S_0 = \alpha M_0 + \beta E_0 + \gamma V_0$

M: top-2 token margin

E: 1 − normalized token entropy

V: CLIP support of $(q \oplus \hat{a}_0)$

(α, β, γ) = (0.5, 0.2, 0.3)

4 Admission gate

admit history?

$S_0 < \tau_S$ *(recent unsure)*

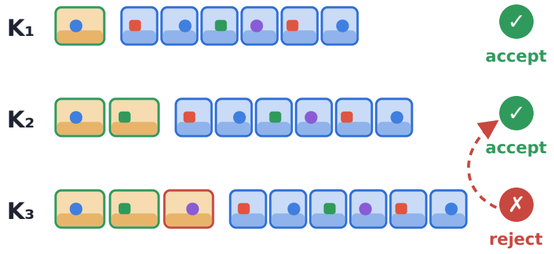
∨

$r_1 \ge \tau_R \wedge h_1$ *supports* $\hat{a}_0$ *less than* R_6 *does*

no → answer $\hat{a}_0$

(identical to Recent-6)

5 Progressive arbitration (B_mem = 3)



prefix rollback: $K^* = K_2$

$A_j = [M_j \ge \tau_M] \cdot [S_{(j-1)} - S_j \le \delta_S] \cdot C_j \cdot P_j$

C_j: a close frame must support the new answer

P_j: answer changes, or S_j rises by ε

6 Answer



$\hat{a} = \hat{a}_{(j^*)}$ from K^*

deepest accepted state; no extra generation

$|K^*| \le 6 + 3$ *frames*

$j^* \in \{0, 1, 2, 3\}$

no memory upkeep: only CLIP image embeddings are cached

exact Recent-6 frames (always kept, hash-identical)

accepted memory state

historical candidate frames (3–30 s band)

rejected state → rollback to last accepted prefix

❄ frozen, training-free: MiniCPM-V-4.6 + CLIP

MCQ and open-ended questions share one controller; all signals come from the generated answer, never from options